

FITBUDDY-AI FITNESS PLAN GENERATOR USING GEMINI MODELS

Phase 4: Project Planning Phase

This phase establishes the development strategy, tools, milestones, and implementation sequence.

Development Environment

A Python virtual environment is created:

```
python -m venv fitbuddy-env
```

The environment is activated and required libraries are installed.

Required packages include:

```
pip install fastapi uvicorn jinja2 sqlalchemy python-multipart google-generativeai
```

Development Plan

The project is divided into major development tasks:

1. Select and integrate the AI model.
2. Configure the FastAPI application.
3. Design the database.
4. Develop workout generation.
5. Develop nutrition tip generation.
6. Implement feedback-based plan updates.

7. Develop frontend templates.
8. Connect frontend and backend.
9. Test the complete application.
10. Prepare local deployment.

Version Control

Git is used to maintain source-code versions and track development changes.

Environment Configuration

Sensitive information such as the Gemini API key is stored using environment variables rather than directly embedding credentials in source code.

Outcome:

A structured development roadmap, environment setup, and task schedule